AUDIO DIGITALIZATION AND SPEECH RECOGNITION FRAMEWORKS IN PYTHON

A Conceptual and Technical Report on Signal Processing, Acoustic and Language Models, Noise Calibration, Deployment Architectures, Reliability, and Audio Formats

Prepared as a technical project report

Focus areas: Python • Digital Signal Processing • Automatic Speech Recognition (ASR) • NLP • Audio Engineering

ABSTRACT

Automatic Speech Recognition (ASR) converts spoken language into written text by combining digital audio capture, signal processing, acoustic or neural representation learning, language modelling, and decoding. A reliable Python speech-to-text system must do more than call a transcription API: it must capture audio at a suitable sampling rate, manage silence and ambient noise, select an appropriate recognition engine, handle failures safely, and preserve the information needed by the model. This report explains the path from an analog sound wave to digital samples and frames, how recognition systems infer linguistic units and words, why signal-to-noise ratio matters, and how cloud services differ from local engines such as Whisper and Vosk. It also distinguishes recognition uncertainty from network failures and discusses the practical effects of sample rate, bit depth, and compression. The report concludes with a reference Python workflow and recommendations for choosing a robust architecture.

Keywords: Automatic Speech Recognition, ASR, digital audio, sampling, SNR, acoustic model, language model, Whisper, Vosk, SpeechRecognition, Python.

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1. INTRODUCTION AND OBJECTIVES

Speech is a natural, fast, and expressive communication channel. A speech-to-text application makes spoken information searchable, accessible, and easier to process. Common applications include voice assistants, meeting transcription, captioning, dictation, call-centre analytics, language learning, and hands-free interfaces.

Automatic Speech Recognition (ASR) is the computational task of estimating a sequence of words from an audio signal. It is closely related to Natural Language Processing (NLP), but it begins with acoustic evidence rather than already-written text. The recognizer must cope with accents, speaking rate, pronunciation variation, room acoustics, background sounds, and ambiguity.

1.1 Objectives

- Explain analog-to-digital conversion and the framing of audio signals.
- Describe how acoustic representations and language context contribute to word recognition.
- Explain noise calibration and the role of signal-to-noise ratio.
- Compare cloud-based recognition with local neural and classical ASR engines.
- Distinguish recognition failures from transport, service, and infrastructure failures.
- Explain the effects of sample rate, bit depth, and compression on transcription quality.
- Provide a practical Python pattern for capturing, transcribing, and handling errors.

1.2 Scope and terminology

This report uses **ASR** for the overall speech-recognition task and **transcription** for its text output. A **frame** is a short segment of audio samples used for analysis. A **phoneme** is a sound category that can distinguish words in a language. Modern end-to-end systems do not always explicitly produce phonemes; they may predict subword tokens or characters directly. Therefore, the phoneme-based explanation below is a useful conceptual model, not a universal description of every modern engine.

2. SYSTEM ARCHITECTURE AND PROCESSING PIPELINE

A practical speech-to-text system can be viewed as a pipeline. Errors introduced early—such as clipping, an incorrect sample rate, or a microphone that is too quiet—can reduce the quality of every later stage.

Stage	Purpose	Typical concerns
1. Capture	Record microphone or load an audio file.	Device selection, permissions, clipping, channel count.
2. Digital conversion	Represent sound as numeric samples.	Sample rate, bit depth, quantization.
3. Pre-processing	Prepare signal for analysis.	Resampling, channel mixing, filtering, level normalization.
4. Segmentation	Identify speech regions or frames.	Silence detection, voice activity detection (VAD), chunk boundaries.
5. Feature/model input	Represent acoustic evidence for the recognizer.	Spectrograms, log-Mel features, learned representations.
6. Decoding	Infer words or tokens from acoustic evidence.	Pronunciation variation, language context, beam search.
7. Post-processing	Format and return text.	Punctuation, capitalization, timestamps, diarization if supported.

2.1 End-to-end flow

```
Microphone / audio file
        |
        v
Analog-to-digital conversion (ADC) or file decoding
        |
        v
PCM samples -> channels / rate / level checks
        |
        v
Pre-processing + voice activity detection
        |
        v
Acoustic features or learned audio representation
        |
        v
ASR model + decoder + language context
        |
        v
Words / punctuation / optional timestamps
        |
        v
Application output, storage, or user interface
```

Not every engine exposes each stage separately. A hosted API may perform most of the pipeline internally, while a local workflow may give the developer control over resampling, VAD, chunking, and decoding parameters.

3. ACOUSTIC AND LANGUAGE MODELING

3.1 From a continuous sound wave to digital samples

Speech begins as changing air pressure. A microphone converts these pressure variations into an electrical signal. An analog-to-digital converter (ADC) measures the signal at discrete time intervals and maps each measurement to a finite numeric value. This produces a sequence of samples that software can store and process.

The **sampling rate** specifies how many samples are captured per second. At 16,000 samples per second, the interval between samples is 1/16,000 second, or 62.5 microseconds. The Nyquist principle states that a sampling rate must exceed twice the highest frequency one wishes to represent, with practical systems also requiring appropriate anti-aliasing filtering. A 16 kHz signal can represent frequencies below approximately 8 kHz, which is generally sufficient for many speech-recognition pipelines.

The **bit depth** describes the number of bits used to represent each sample in uncompressed PCM audio. More bits allow finer amplitude quantization and a wider theoretical dynamic range. For example, 16-bit PCM provides 65,536 possible sample codes. In real recordings, microphone noise, room noise, and recording gain often matter more than simply increasing bit depth.

3.2 Framing and windowing

Speech is not stationary over long periods: vowels, consonants, pauses, and transitions change rapidly. Systems therefore analyse short overlapping frames, often around 20–30 milliseconds long, with a smaller hop between frames. For a 25 ms frame at 16 kHz, there are 400 samples. A window function, such as a Hann window, reduces abrupt boundary effects before frequency analysis.

The short-time Fourier transform (STFT) estimates how signal energy is distributed across frequencies over time. Many recognizers use log-Mel spectrograms or related learned features because they provide a compact representation of speech-relevant spectral patterns. Neural systems may also learn representations directly from waveform samples.

3.3 Acoustic modeling

An acoustic model estimates which linguistic units are supported by the audio. Traditional systems commonly combined acoustic models with pronunciation dictionaries and language models. More recent systems may use neural encoders and decoders trained end-to-end. Depending on the architecture, intermediate units can be phonetic states, phonemes, characters, word pieces, or other tokens.

A simplified conceptual sequence is:

```
Audio samples -> frames/features -> acoustic evidence
              -> phonemes or subword tokens
              -> candidate word sequence
              -> decoded text
```

For example, the acoustic evidence in a short recording may be compatible with several similar-sounding words. The recognizer uses learned sound patterns and context to rank possible outputs. The output is an estimate, not a perfect measurement of what was said.

3.4 Language modeling and decoding

A language model represents patterns in word or token sequences. It can favour a grammatically or contextually plausible sequence when the acoustics are ambiguous. In a classic formulation, a decoder seeks a word sequence W that best explains audio observations A, often described conceptually as maximizing P(W|A). Using Bayes' rule, a classic hybrid system may combine an acoustic likelihood P(A|W) with a language prior P(W). Modern end-to-end systems often learn these relationships jointly.

Context is helpful but can also introduce errors. A model may replace an unfamiliar name, technical term, or uncommon phrase with a more common phrase that sounds similar. Domain vocabulary, custom prompts where supported, and careful review can improve results, but they do not guarantee correctness.

3.5 Why recognition is probabilistic

- Different speakers pronounce the same word differently.
- Coarticulation causes neighbouring sounds to influence one another.
- Accents, microphone placement, reverberation, and speaking speed alter the signal.
- Words can sound alike, and the surrounding sentence may be needed to disambiguate them.
- The model may not have seen a rare name or specialist term often enough during training.

4. AMBIENT NOISE CALIBRATION AND SIGNAL-TO-NOISE RATIO

4.1 Definition of SNR

Signal-to-noise ratio (SNR) compares the strength of the desired speech signal with unwanted background sound. For power measurements, SNR in decibels is commonly expressed as 10 log10(Psignal/Pnoise). For RMS amplitude measurements, the equivalent expression is 20 log10(Asignal/Anoise), assuming compatible measurement conditions. A higher SNR generally means speech is easier to distinguish from noise, but SNR alone does not fully describe speech quality: reverberation, overlapping speakers, clipping, and the type of noise also matter.

4.2 Why calibration matters

Automatic gain control and noise calibration attempt to make speech usable without amplifying background noise excessively. In the Python SpeechRecognition library, a common pattern is to call `adjust_for_ambient_noise()` while the user is quiet. The library estimates an energy threshold from the observed ambient sound and uses it to help decide when speech begins and ends.

This calibration is not the same as removing all noise from the waveform. It mainly helps the library's energy-based phrase detection decide which sound levels should count as speech. If a person speaks during calibration, the estimated threshold may become unsuitable. If the room noise changes later, the original threshold may also become stale.

4.3 Thresholding, VAD, and denoising

A voice activity detector (VAD) estimates which regions contain speech. A simple energy threshold is inexpensive but can mistake a loud fan for speech or miss quiet speech. More advanced VAD systems use spectral or neural cues. Denoising algorithms attempt to suppress unwanted components while preserving speech; aggressive denoising can distort consonants and other useful acoustic cues.

- **Calibrate during a short quiet interval:** do not speak while the ambient level is measured.
- **Keep microphone distance consistent:** large changes in distance change speech level and room-noise balance.
- **Avoid clipping:** if the waveform repeatedly reaches its maximum or minimum value, information has already been lost.
- **Use VAD thoughtfully:** speech segmentation can reduce wasted processing but must not cut off the start or end of words.
- **Test in realistic environments:** evaluate with the actual room, microphone, speaker distance, and background sounds.

4.4 Noise handling is not a universal fix

A recognizer may still produce incorrect text in a noisy environment even if the application successfully detects a phrase. Recognition accuracy depends on the model's training data, the noise type, microphone quality, language, and speech clarity. For difficult recordings, improving capture conditions is often more effective than adding many layers of aggressive filtering.

5. CLOUD API VS. LOCAL PROCESSING

Cloud-based recognition sends audio to a remote service for processing. Local recognition runs the model on the user's device or an organisation-controlled machine. The best choice depends on privacy, connectivity, hardware, language support, cost, latency, and operational constraints.

Dimension	Cloud service (e.g., Google Web Speech API)	Local engine (e.g., Whisper or Vosk)
Processing location	Audio is sent to a provider service; exact handling depends on the API and its terms.	Audio can remain on the device or controlled server.
Connectivity	Usually requires network access and a reachable service.	Can operate offline after model installation.
Latency	Includes upload, network round-trip, queueing, and server inference.	Avoids network delay; model speed depends on hardware and size.
Privacy	Requires reviewing provider retention, logging, region, and contractual policies.	Offers more direct control over storage and access; local logs still need protection.
Accuracy	May benefit from powerful managed models and service-side updates.	Varies with model size, language, hardware, and acoustic conditions.
Maintenance	Provider manages model deployment; API behaviour and limits can change.	Developer manages model files, dependencies, upgrades, and performance.
Cost	May involve quotas, billing, usage limits, or availability constraints.	No per-request cloud fee, but hardware, energy, and maintenance have costs.

Dimension	Cloud service (e.g., Google Web Speech API)	Local engine (e.g., Whisper or Vosk)
Reliability	Dependent on network, credentials, service availability, and quotas.	Dependent on device resources, installation, and local system health.

5.1 Google Web Speech API through SpeechRecognition

The Python `SpeechRecognition` package offers a convenient interface to several recognizers. Its Google Web Speech integration is often used for prototypes and demonstrations. Availability, authentication requirements, quotas, terms, and supported options can vary; a wrapper method should not be assumed to provide the same service guarantees as a production cloud speech API.

5.2 OpenAI Whisper

Whisper is a neural speech-recognition model family that can be run locally through compatible implementations. It supports multilingual transcription and, for supported tasks, translation into English. Larger models generally require more compute and memory, while smaller models may run faster but can be less accurate on challenging speech. Actual performance depends on hardware, model variant, decoding settings, language, and audio quality.

5.3 Vosk

Vosk is an offline speech-recognition toolkit with models for a range of languages and deployment environments. It can be useful for streaming and resource-constrained applications. Model size, language coverage, vocabulary, and accuracy vary by model. A smaller local model may be attractive for low-latency offline use, but it should be benchmarked on representative recordings before deployment.

5.4 Architectural decision guide

- **Choose cloud processing** when managed infrastructure, rapid integration, or a provider's language and model features outweigh network and data-governance concerns.
- **Choose local processing** when offline operation, control over audio retention, or predictable network independence is essential.
- **Choose a hybrid architecture** when a local first-pass recognizer can handle routine audio and a cloud service is available as an optional fallback under an explicit privacy policy.

No engine is universally the most accurate. Compare candidates using the same audio, language, recording conditions, and evaluation metric rather than relying on broad claims.

6. EXCEPTION HANDLING AND STREAM RELIABILITY

6.1 Recognition errors versus service errors

Robust code distinguishes the recognizer's inability to interpret the audio from the inability to complete the request. These are different failure classes and should lead to different user messages and recovery strategies.

Failure	Meaning	Typical response
UnknownValueError	The recognizer could not confidently interpret the supplied audio as speech.	Ask for a clearer recording, check volume/noise, or offer manual correction.
RequestError	The request to an online recognition service failed—for example, due to connectivity, service, or request problems.	Check connection, endpoint/API configuration, quotas, and service status; retry cautiously.
WaitTimeoutError	A phrase was not detected within the configured listening timeout.	Prompt the user to speak and adjust timeout or phrase limits if appropriate.

Failure	Meaning	Typical response
OSError / device errors	Audio device, file, driver, or input stream operation failed.	Check microphone permissions, device selection, file path, and supported format.
Model/runtime errors	A local model could not load or execute due to missing files, incompatible dependencies, or resource limits.	Validate installation, model path, memory, and runtime compatibility.

In the SpeechRecognition package, `UnknownValueError` and `RequestError` are commonly associated with recognizer calls. Their exact applicability depends on the recognizer method used. A local offline engine will not usually raise a cloud-request error for inference because it does not make that request.

6.2 Safe retry strategy

- Retry transient network failures with a small retry limit and exponential backoff.
- Do not endlessly retry an audio clip that the recognizer consistently cannot understand.
- Keep timeouts finite so a stalled service or microphone does not block the application indefinitely.
- Log error categories and timing, but avoid logging raw audio or sensitive transcripts unless necessary and authorised.
- Give the user actionable messages without exposing API keys, internal endpoints, or private diagnostic data.

6.3 Stream reliability

For continuous or long recordings, process audio in chunks and preserve their order. Chunking reduces memory use and can lower perceived delay, but a chunk that begins or ends inside a word may harm recognition. Overlap can preserve boundary context, although duplicated words may then need reconciliation. Streaming systems should track chunk timestamps, detect dropped audio, and define what happens when the network or model becomes unavailable.

6.4 Example exception handling

```
import speech_recognition as sr

recognizer = sr.Recognizer()

try:
    with sr.Microphone() as source:
        print("Please speak after the short calibration period.")
        recognizer.adjust_for_ambient_noise(source, duration=0.8)
        audio = recognizer.listen(
            source,
            timeout=5,          # wait for speech to begin
            phrase_time_limit=12 # limit one phrase
        )

    text = recognizer.recognize_google(audio)
    print("Transcript:", text)

except sr.WaitTimeoutError:
    print("No speech detected before the timeout.")
except sr.UnknownValueError:
    print("Speech was detected, but the words could not be understood.")
except sr.RequestError as exc:
    print("The recognition service request failed:", exc)
except OSError as exc:
    print("Microphone or audio-device problem:", exc)
```

The example is intended as a starting point. The microphone requires suitable operating-system permissions and audio dependencies. In production, handle configuration, logging, cancellation, and user consent explicitly.

7. AUDIO FORMAT, SAMPLING RATE, BIT DEPTH, AND CODECS

7.1 Sampling rate: 16 kHz versus 44.1 kHz

Speech models commonly expect a specific input rate, often 16 kHz for many ASR pipelines, though requirements vary. A 44.1 kHz recording contains more samples per second and can preserve a wider frequency range, but that does not automatically make transcription more accurate. If a model expects 16 kHz, the audio should be properly resampled to 16 kHz rather than merely relabelled with a different rate.

Incorrectly changing metadata without converting the samples changes the interpreted duration and pitch. Proper resampling uses interpolation and filtering to create a new sample sequence at the target rate. When downsampling, anti-aliasing filtering is needed to prevent high-frequency content from folding into lower frequencies.

7.2 Bit depth and dynamic range

Bit depth determines the resolution of PCM amplitude measurements. Higher bit depth reduces quantization error and can preserve quiet details when the recording chain has sufficient quality. However, it cannot restore speech lost through clipping, severe background noise, or a poor microphone. For many speech workflows, clean 16-bit PCM is a practical interchange format.

7.3 WAV versus MP3

WAV is a container, not a guarantee of uncompressed audio: it often contains PCM samples, but other encodings are possible. MP3 is a lossy compression format that removes information judged less perceptually important. MP3 can reduce storage and transfer costs, but compression artefacts may affect consonants, fricatives, and subtle speech cues, especially at low bit rates or after repeated encoding.

Property	WAV with PCM	MP3
Compression	Usually uncompressed in common speech datasets.	Lossy compression.
File size	Larger.	Usually much smaller.
Processing	Straightforward for many audio libraries and model pipelines.	May need decoding to PCM before processing.
Recognition impact	A good choice for clean reference audio.	Often adequate at reasonable quality, but artefacts can hurt difficult recordings.
Important caveat	Check the actual codec, sample rate, channels, and bit depth.	Check bitrate and generation quality; converting MP3 to WAV does not restore lost data.

7.4 Mono and stereo

Many recognizers accept mono audio or perform best with a single channel. Stereo recordings can contain useful spatial information, but they may also have different speakers or noise levels in each channel. Downmixing should be done carefully: phase cancellation or an especially noisy channel can reduce clarity. If separate speakers are recorded on separate channels, retaining channels may be more useful than immediately mixing them.

7.5 Format validation checklist

- Confirm the recognizer's accepted formats, sample rates, channel count, and maximum input duration.
- Decode compressed files to a supported PCM representation when required.
- Resample with an audio library; do not simply change the sample-rate label.

- Check for silence, clipping, excessive gain, and very low recording levels.
- Preserve the original file when traceability or reprocessing may be needed.

8. PYTHON IMPLEMENTATION PATTERNS

8.1 Basic architecture

A maintainable application separates capture, audio preparation, recognition, error handling, and output formatting. This makes it easier to replace the recognition engine without rewriting the microphone interface.

```
def transcribe_audio(audio, recognizer, engine="google"):
    if engine == "google":
        return recognizer.recognize_google(audio)
    raise ValueError(f"Unsupported engine: {engine}")
```

For larger projects, define a common interface such as `transcribe(audio) -> TranscriptResult`. Each engine adapter can return text, language, timestamps if available, and structured errors. This keeps provider-specific details out of the rest of the application.

8.2 Offline processing concept

```
# Conceptual example: Vosk accepts PCM audio through its recognizer.
# The audio must be converted to the sample rate expected by the model.
from vosk import Model, KaldiRecognizer
import json
import wave

model = Model("path/to/vosk-model")
with wave.open("speech.wav", "rb") as wf:
    if wf.getnchannels() != 1:
        raise ValueError("Convert audio to mono first")
    if wf.getsampwidth() != 2:
        raise ValueError("Expected 16-bit PCM audio")

    recognizer = KaldiRecognizer(model, wf.getframerate())
    while True:
        data = wf.readframes(4000)
        if not data:
            break
        recognizer.AcceptWaveform(data)

    result = json.loads(recognizer.FinalResult())
    print(result.get("text", ""))
```

This is a conceptual Vosk example: the chosen model, audio format, language, and sample rate must be compatible. A real application should also validate compression, handle partial results, and report model-loading errors.

8.3 Operational recommendations

- Make engine selection configurable instead of hard-coding one provider.
- Use environment variables or a secret manager for credentials; never commit keys to source control.
- Set maximum audio duration and file size to control memory and cost.
- Return structured results containing transcript, duration, engine, and error status.
- Keep a human-review path for names, numbers, legal content, and other high-impact transcription.

9. TESTING, EVALUATION, AND PERFORMANCE METRICS

9.1 Word Error Rate

Word Error Rate (WER) is a widely used ASR metric. It compares a hypothesis transcript with a reference transcript using substitutions (S), deletions (D), and insertions (I):

```
WER = (S + D + I) / N
```

Here, N is the number of words in the reference transcript. Lower WER is better. WER can exceed 1.0 when insertions are numerous. Evaluation conventions for punctuation, case, numbers, and tokenisation should be defined consistently.

9.2 Complementary metrics

- **Character Error Rate (CER):** useful for languages or tasks where character-level differences matter.
- **Latency:** elapsed time between the end of speech and transcript availability, or end-to-end delay for streaming.
- **Real-time factor (RTF):** processing time divided by audio duration; below 1 means processing is faster than real time under that test.
- **Robustness:** quality across speakers, accents, noise types, distances, and recording devices.
- **Operational reliability:** request success rate, timeout rate, crash rate, and recovery time.

9.3 Recommended test matrix

Test dimension	Example cases
Environment	Quiet room, fan noise, traffic, reverberant room.
Speaker	Different accents, speaking rates, vocal ranges, and distances.
Content	Short commands, long paragraphs, names, numbers, technical vocabulary.
Signal	16 kHz PCM, 44.1 kHz source resampled correctly, compressed audio, clipped audio.
Failure modes	No speech, intermittent network, quota limit, unavailable microphone, missing model.
Output	WER/CER, latency, completeness, punctuation, timestamps where supported.

A fair comparison should use a fixed, representative evaluation set and document model versions, settings, and hardware. Avoid tuning repeatedly against the final test set, because this can produce an overly optimistic estimate of real-world performance.

10. SECURITY, ETHICS, AND OPERATIONAL CONSIDERATIONS

- **Consent:** tell speakers when audio is being recorded or transcribed, especially in meetings and shared spaces.
- **Data minimisation:** collect and retain only the audio and transcript information needed for the task.
- **Access control:** protect recordings, transcripts, model logs, and API credentials.
- **Provider review:** understand the selected service's retention, processing location, training-use terms, and deletion options.
- **Bias and fairness:** test performance across accents, dialects, ages, and speaking styles relevant to the intended users.
- **Human oversight:** treat machine transcripts as fallible, particularly for medical, legal, educational, financial, or safety-critical use.

Offline processing can reduce the need to transmit audio to a third party, but it does not automatically make a system secure. Local files, temporary recordings, backups, and application logs can still expose sensitive information.

11. CHALLENGES AND FUTURE DIRECTIONS

Speech-recognition research continues to improve multilingual support, streaming performance, robustness to noise, speaker separation, and adaptation to specialised vocabularies. Several challenges remain:

- **Overlapping speakers:** ordinary ASR may merge words or miss a speaker when voices overlap; diarization and source separation can help but add complexity.
- **Code-switching:** speakers may switch languages within a sentence, requiring suitable multilingual models and evaluation.
- **Domain-specific words:** uncommon names, acronyms, and technical terms may be replaced by more frequent words.
- **Edge deployment:** small models reduce memory and latency but require careful evaluation of accuracy trade-offs.
- **Streaming context:** low-latency systems must balance early partial output against the benefit of hearing more context.
- **Trustworthy output:** confidence indicators, timestamps, provenance, and human correction workflows can make transcripts easier to verify.

A practical future architecture is modular: capture high-quality audio, use VAD to segment speech, run a model appropriate to the privacy and latency requirements, preserve timestamps and confidence information when available, and route uncertain or high-impact content for human review.

12. CONCLUSION

Audio digitalization and speech recognition form a connected engineering pipeline. The analog signal must be sampled and represented correctly; audio frames and learned acoustic features must retain the cues needed to distinguish speech; and a decoder must infer a likely sequence of words from imperfect evidence. Ambient-noise calibration helps determine when speech is present, while SNR, microphone quality, clipping, reverberation, and segmentation all affect transcription reliability.

Cloud recognition can provide convenient managed processing, but it introduces network dependency and requires careful data-governance review. Local systems such as Whisper and Vosk can support offline operation and stronger control over audio handling, but they require model management and hardware-aware testing. Reliable Python applications should separate recognition uncertainty from network and device failures, validate audio formats rather than trusting file extensions, and evaluate results using consistent metrics such as WER and latency.

Ultimately, the most accurate and reliable system is not selected by model name alone. It is the system that has been tested with the target language, speakers, devices, acoustic conditions, privacy requirements, and deployment constraints.

APPENDIX A. PRACTICAL CHECKLIST

- ■ Microphone permission and device selection have been verified.
- ■ Ambient calibration occurs while the room is quiet.
- ■ Audio is not clipped and the speaking distance is reasonable.
- ■ Sample rate is converted correctly to the model's expected rate.

- ■ Channel count, bit depth, codec, and duration are validated.
- ■ The application handles timeout, unintelligible speech, request, and device errors separately.
- ■ Retries are bounded and do not create duplicate actions or costs.
- ■ Audio retention and transcript access follow an explicit privacy policy.
- ■ The selected engine is evaluated on representative recordings.
- ■ Users can correct important transcripts and report recognition failures.

REFERENCES AND FURTHER READING

1. Python SpeechRecognition project documentation and source code: https://github.com/Uberi/speech_recognition

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7. Oppenheim, A. V. and Schafer, R. W., Discrete-Time Signal Processing, Pearson.

Note: Library behaviour, API requirements, model availability, and provider policies can change. Consult the official documentation for the exact versions used in an implementation.